

SHELIKHOVA, Russian Federation

WMO: 259220

Lat: 60.717N

Lon: 155.883E

Elev: 33

StdP: 100.93

Time Zone: 11.00 (E11)

Period: 88-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-25.4	-33.2	0.2	-26.5	-31.6	0.2	-24.9	20.4	-8.9	17.4	-6.9	5.7	290	0.897

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.7	18.0	13.7	16.6	13.0	15.4	12.7	14.7	16.5	14.0	15.5	13.4	14.8	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	15.3	13.2	9.5	14.6	12.6	9.1	14.0	41.3	16.6	39.4	15.6	37.8	14.9	18.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.1	15.4	13.1	DB	-29.0	22.8	2.2	2.7	-30.6	24.8	-31.9	26.3	-33.1	27.8	-34.7	29.8	
			WB	-29.1	15.9	1.9	0.7	-30.5	16.4	-31.6	16.8	-32.7	17.2	-34.1	17.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-15.3	-15.2	-11.7	-4.6	2.1	8.7	12.6	12.3	7.7	-0.2	-7.9	-12.6
	DBStd	11.07	6.20	5.96	5.77	3.92	2.91	2.79	2.03	1.86	2.83	3.77	5.26	5.42
	HDD10.0	4538	786	704	671	437	245	57	1	4	78	317	538	700
	HDD18.3	7394	1044	938	929	687	503	288	177	187	318	575	788	958
	CDD10.0	186	0	0	0	0	1	18	83	75	9	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	6.7	6.3	6.4	5.4	4.2	3.9	3.9	4.3	5.0	6.3	7.9	8.0
Precipitation	PrecAvg	468	15	16	25	24	21	29	45	91	72	66	42	18
	PrecMax	640	33	39	66	122	63	74	116	253	189	147	161	65
	PrecMin	303	0	0	2	2	2	1	2	10	10	12	0	2
	PrecStd	104	10	11	18	26	16	22	33	54	50	33	36	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	0.8	-0.8	4.5	13.6	20.1	22.1	18.6	15.7	9.1	3.6	0.0
		MCWB	-0.5	0.1	-2.5	1.1	7.8	12.5	15.0	13.9	11.5	6.8	2.5	-0.6
	2%	DB	-2.2	-1.6	-1.9	2.6	9.7	15.9	19.2	17.2	13.6	8.1	2.4	-1.7
		MCWB	-2.7	-2.1	-2.8	0.4	6.1	11.2	14.2	13.4	10.8	6.2	1.6	-2.6
	5%	DB	-4.1	-4.4	-2.7	1.0	7.7	14.2	17.2	16.1	12.5	6.7	1.1	-3.4
		MCWB	-4.9	-5.2	-3.5	-1.0	4.7	10.6	13.7	13.1	10.2	4.5	0.5	-4.3
	10%	DB	-6.4	-6.5	-4.0	0.1	6.4	13.0	15.7	15.2	11.6	5.1	-0.6	-5.0
		MCWB	-7.4	-7.4	-5.0	-1.1	3.9	10.2	13.0	12.8	9.5	2.8	-1.3	-5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	0.2	-1.9	1.5	7.9	13.6	16.0	15.6	13.2	7.8	2.7	-0.4
		MCDB	-0.3	0.7	-1.5	3.5	12.0	17.1	19.5	16.9	14.2	8.6	3.4	0.1
	2%	WB	-2.5	-2.0	-2.5	0.6	6.3	12.1	15.0	14.6	12.0	6.7	1.7	-2.5
		MCDB	-2.2	-1.6	-2.0	2.1	9.2	14.3	17.5	15.7	12.8	7.7	2.4	-1.9
	5%	WB	-4.8	-5.0	-3.3	-0.3	5.2	11.3	14.2	14.0	10.9	5.0	0.6	-4.2
		MCDB	-4.2	-4.5	-2.7	0.4	7.4	13.4	16.2	15.1	11.8	6.3	1.0	-3.5
	10%	WB	-7.4	-7.3	-4.8	-0.9	4.2	10.6	13.6	13.4	10.1	3.1	-1.2	-5.9
		MCDB	-6.4	-6.4	-4.2	-0.2	5.9	12.6	15.2	14.7	11.2	4.6	-0.8	-5.0
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.4	5.1	4.6	4.3	4.4	3.7	4.1	4.8	5.1	4.5	4.2
		MCDBR	4.7	3.6	3.9	4.0	6.6	7.1	7.3	5.6	5.1	5.1	3.5	5.2
		MCWBR	4.7	3.6	3.4	3.2	4.3	4.7	4.4	3.8	3.8	4.2	3.4	5.0
	5% WB	MCDBR	4.4	4.0	3.2	3.6	6.0	5.9	5.2	4.4	3.5	4.6	3.6	5.1
		MCWBR	4.6	4.2	3.1	2.9	4.1	4.3	3.5	3.4	3.0	4.3	3.4	4.9
Clear-Sky Solar Irradiance	taub	0.219	0.265	0.321	0.393	0.423	0.401	0.455	0.425	0.336	0.309	0.253	0.208	
	taud	2.047	2.091	1.978	1.853	1.988	2.290	2.178	2.249	2.481	2.277	2.099	1.976	
	Ebn at Noon	533	714	774	778	792	827	764	756	788	659	467	366	
	Edh at Noon	41	75	119	165	156	118	130	112	74	63	39	27	
All-Sky Solar Radiation	RadAvg	0.39	1.30	3.06	4.93	5.36	5.29	4.97	3.67	2.51	1.28	0.44	0.17	
	RadStd	0.04	0.09	0.19	0.34	0.50	0.55	0.41	0.43	0.34	0.15	0.04	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page